

Musavvir Haque

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🎓 Google Scholar | 🆔 0009-0008-0521-3474 | 🌐 Portfolio | 👤 IEEE Collabratec | 📄 ResearchGate

Research Profile

Aspiring PhD researcher in **computational materials science** and **materials**, with hands-on experience in large-scale molecular dynamics (MD) simulation and machine learning (ML) surrogates for materials discovery. Primary research focus: **plasma-material interactions in fusion-relevant tungsten** (hydrogen/isotope retention, radiation damage, oxide dynamics) and **ML-accelerated ion implantation modelling** in semiconductor substrates. Proficient in LAMMPS, XGBoost/Random Forest pipelines, and physics-aware surrogate design. One first authored manuscript under review; two accepted IEEE conference papers.

Research Interests

Plasma-Facing Materials for Fusion Reactors (ITER/DEMO) · Hydrogen/Tritium Retention & Radiation Damage in Tungsten · ML/AI Surrogates for DFT & MD · Ion Implantation Physics · Materials Degradation · High-Performance Atomistic Simulation (LAMMPS)

Publications

Journal Articles (Under Review)

[J1] Musavvir Haque, Md. Amzad Hossain, “Hydrogen Implantation and Diffusion in BCC Tungsten: A Systematic Molecular Dynamics Study over 5–300 eV and 300–1500 K,” *Fusion Engineering and Design*, Under Review (MS# FUSENGDES-D-26-00360, Submitted May 2026).

1,000 classical MD simulations in LAMMPS; resolves non-monotonic H retention, Arrhenius diffusion parameters, and cascade-dominated regimes across a 10×10 energy–temperature parameter space in BCC tungsten.

[J2] A. S. Fahim, **M. Haque**, A. Taher, N. M. Sathi, J. Islam, Md. M. Hasan and Md. A. Hossain, “A Gatekeeper-Constrained Dual-Stage Random Forest Surrogate for Multi-Output Ion Implantation Prediction Across Si, SiC, and GaAs,” *Computational Materials Science* (Elsevier), Manuscript No. COMMAT-D-26-02010R1. [Under Review – Revision Submitted May 2026]

B.Sc. Thesis

[T1] **Multi-Scale DFT-MD-ML Modelling on Hydrogen Isotope Retention and Oxide Dynamics in Tungsten with Plasma Exposure for Fusion Reactor**

B.Sc. Thesis, Jashore University of Science and Technology

Supervisor: Md. Tariquzzaman, Asst. Prof.

Multiscale framework integrating DFT/MD-derived physical trends with ML for hydrogen isotope retention and oxide-layer evolution in plasma-exposed tungsten. Synthetic datasets emulate experimentally realistic plasma-material interaction behaviour. **XGBoost**: MAE = 0.017 eV, $R^2 = 0.88$ (binding energy); MAE = 0.91%, $R^2 = 0.87$ (crack density). Validated against TDS experiments and WEST tokamak observations.

Conference Papers

[C1] M. Haque, R. I. Rony, S. Mustak and I. Khalil, “Adaptive Multi-Scale Variational Mode Decomposition with Transfer Entropy for Real-Time Detection of Arrhythmia Transitions in Wearable ECG Devices,” *Proc. 5th Int. Conf. on Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, 2026, pp. 1–6.
doi:10.1109/ICECTE69292.2026.11429390

[C2] M. Haque *et al.*, “Secure Academic Assessment Using IoT Based Dual Mode Biometric Attendance System for University,” *2025 Int. Conf. on Quantum Photonics, AI, and Networking (QPAIN)*, Rangpur, Jul. 2025.
doi:10.1109/QPAIN66474.2025.11171806

[C3] N. M. Sathi, Md. M. H. Rocky, N. R. Ritu, Md. S. Hossain, I. A. Mim, Md. M. Rahman, **M. Haque**, A. S. Fahim, Md. R. Khan and Md. A. Hossain, “A DFT Study on Penta-Boron Phosphorus (P-BP₅) Nanosheet for Oxygen-Based Toxic Gases Sensing,” *2026 2nd Int. Conf. on Engineering and Technology for Sustainable Development (ETSD)*, 2026. [Accepted, Paper ID: 136]

Education

B.Sc. (Engg.) in Electrical and Electronic Engineering

CGPA: 3.47 / 4.00

Jashore University of Science and Technology (JUST), Bangladesh

- Relevant coursework: VLSI, Electromagnetic Theory, Digital Signal Processing, Numerical Methods, Control System
- Thesis: Multi-Scale DFT–MD–ML Modelling on Hydrogen Isotope Retention in Fusion-Relevant Tungsten

Higher Secondary Certificate (Science), Notre Dame College

GPA: 5.00 / 5.00

Research & Computational Skills

- **Atomistic Simulation:** LAMMPS (large-scale MD, GPAW, ASE, pymatgen, EAM/alloy potentials, NPT/NVT ensembles, cascade simulations); exposure to DFT concepts (VASP/Quantum ESPRESSO principles)
- **Machine Learning for Materials:** XGBoost, Random Forest, physics-informed feature engineering, surrogate modelling, regression pipelines (Python: scikit-learn, pandas, NumPy, Matplotlib)
- **Ion Implantation / Plasma Physics:** SRIM/TRIM simulations, implantation depth profiling, hydrogen retention analysis, radiation damage metrics (DPA, NIEL)
- **Signal Processing:** Variational Mode Decomposition, Transfer Entropy, real-time wearable biosignal processing (ECG), MATLAB
- **Engineering Tools:** SOLIDWORKS, Proteus, ModelSim, PVsyst, PLC (TIA Portal), PCB Design
- **Programming:** Python, MATLAB, Embedded C, Arduino
- **Scientific Writing:** $\LaTeX$, IEEE manuscript preparation, technical reporting

Academic Leadership & IEEE Activities

Student Researcher

ILAMPS, JUST

- Student researcher at the Interdisciplinary Laboratory of Advanced Materials Processing and Simulation (ILAMPS), conducting MD/DFT-based computational modelling and ML-driven surrogate development for plasma–material interaction studies in fusion-relevant materials.

Chair, IEEE Signal Processing Society JUST Student Branch Chapter

- Elected Chair of IEEE SPS JUST; leading technical workshops, research mentorship, and student engagement in signal processing and computational sciences.
- Organised the *SigMa Mentorship Programme 2026* fostering undergraduate research culture.

Vice Chair (Technical), IEEE JUST Student Branch

- Organised seminars, workshops, and an international conference (ICETSD 2025); promoted research and innovation across EEE and CS disciplines.

Guest Speaker – Physical Design (PNR), EEE Club JUST

- Delivered invited seminar on Placement & Routing following industrial training at Ulkasemi Ltd.; covered ASIC design, EDA tools, and semiconductor career pathways.

Speaker – Embedded AI / Speech & Audio Processing for Wearables, IEEE SPS DeKUT Student Branch Chapter

- Delivered a webinar session on embedded AI, speech, and audio processing for next-generation wearable devices.
- Organised by IEEE Signal Processing Society, DeKUT SBC in collaboration with IEEE CAS & IEEE DeKUT SB.

Judge – SignalQuest 2026 (IEEE SPS Day Quiz Competition), IEEE SPS JUST Student Branch Chapter

- Served as a judge for the IEEE SPS Day Quiz Competition (SignalQuest 2026), organized by the IEEE SPS JUST Student Branch Chapter.

Honours & Awards

- **1st Place – Poster Presentation**, Sustainable IoT-Based Biometric Attendance System, JUST, May 2025
- **1st Place – Best Paper Presentation**, 2nd Int. Conf. on Engineering and Technology for Sustainable Development (ETSD 2026), Jashore University of Science and Technology, May 2026 — for “Integrated DFT–MD–ML Multiscale Modelling of Hydrogen Isotope Trapping and Oxide Evolution in Tungsten under Fusion-Relevant Plasma Exposure”

Relevant Training & Certifications

- **Machine Learning With Python** – Bangladesh Computer Council (EDGE Project), JUST, Sep 2025 (*ID: EDGE-DSTS-115-6728-00001*)
- **EF SET English Proficiency: C2 Proficient** (82/100) – cert.efset.org/C2S2MJ, Feb 2026
- **IEEE Signal Processing Society Member** (Jan 2025 – present); **IEEE Power and Energy Society Member**
- **Solar Training: PV Technology & System Design** – Daffodil International University, Jan 2026 (*ID: RE26312*)
- **Industrial Training – Electrical Engineering & Instrumentation** (PLC, sensors, automation), TICI, Jul 2025
- **Introduction to IoT** – Cisco, Jul 2023; **SOLIDWORKS Associate (CSWA) Course** – Udemy, Jul 2023

Select Research-Aligned Projects

IoT-Based Dual-Mode Biometric Attendance System (ESP32 + Supabase)

AES-256 encrypted, role-based cloud attendance system with offline sync; awarded 1st place at JUST poster competition and accepted at IEEE QPAIN 2025.

Energy Audit – Saleha Metal Industries Ltd. (5.1M kWh/yr, 400 kVA transformer)

Comprehensive industrial energy audit; recommendations for load sharing optimisation and renewable energy integration.

Industrial Visit – Bheramara 410 MW Combined Cycle Power Plant (CCPP), IEEE JUST SB

Participated in an industrial visit on **October 12, 2023**, gaining practical exposure to large-scale power plant operations, combined-cycle power generation, and modern energy technologies.

References

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